LAB EXPERIMENT 2

Experiment 2: Finite State Automaton for Strings Ending with 'ab'

Aim

To implement a basic finite state automaton that accepts strings ending with the pattern 'ab'.

Algorithm

1. Start the program.
2. Read a string containing the symbols a and b.
3. Initialize the automaton to state q0.
4. For each input character, change state according to the transition rules.
5. Use q2 as the accepting state because it represents a string ending in 'ab'.
6. If the final state is q2, display Accepted; otherwise, display Rejected.
7. Stop the program.

Code

```
def check_string(s):
    # q0 = 0, q1 = 1, q2 = 2 (accepting state)
    state = 0

    for ch in s:
        if state == 0:
            if ch == 'a':
                state = 1
            else:
                state = 0
        elif state == 1:
            if ch == 'a':
                state = 1
            else:  # ch == 'b'
                state = 2
        else:  # state == 2
            if ch == 'a':
                state = 1
            else:  # ch == 'b'
                state = 0

    if state == 2:
        print("Accepted: String ends with 'ab'")
    else:
        print("Rejected: String does not end with 'ab'")

s = input("Enter a string containing a and b: ")
if all(ch in "ab" for ch in s):
    check_string(s)
else:
    print("Invalid input: use only a and b.")
```

Output / Screenshot

```
s = input("Enter a string: ")
state = 0

for ch in s:
    if state == 0:
        if ch == 'a':
            state = 1
    elif state == 1:
        if ch == 'a':
            state = 1
        elif ch == 'b':
            state = 2
    elif state == 2:
        if ch == 'a':
            state = 1
        elif ch == 'b':
            state = 0

if state == 2:
    print("String Accepted")
else:
    print("String Rejected")
```

```
Enter a string: abab
String Accepted
```

Result

Thus, the finite state automaton was implemented and tested to accept strings that end with 'ab' and reject strings that do not.